MILITARY STANDARD

MIL-STD-1320-216 (NAVY)

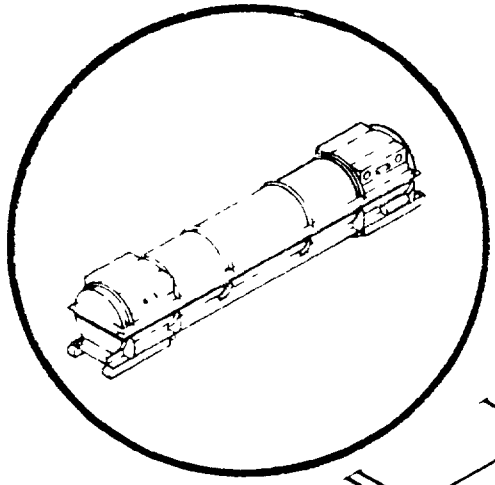
1ST SEPTEMBER 1985

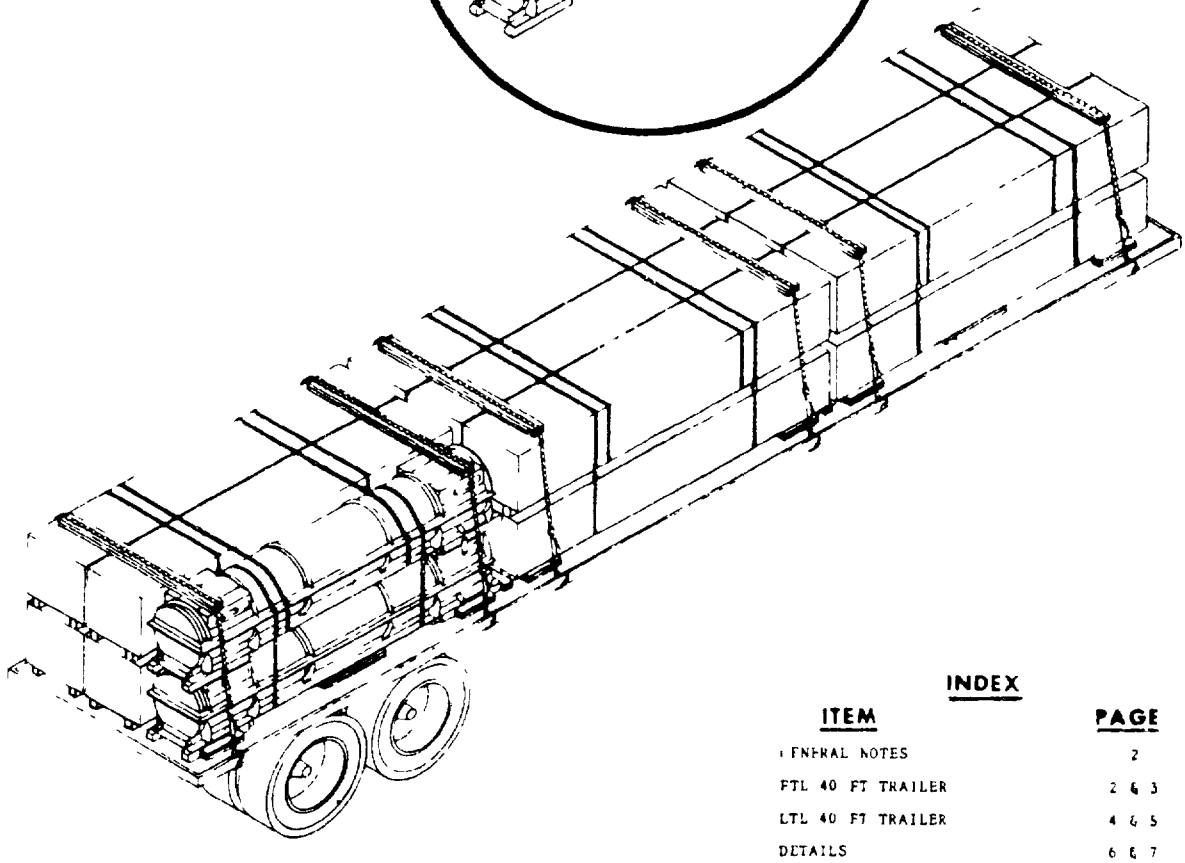
TRUCKLOADING
MINE, UNDERWATER MK 67 MOD 2 (SLMM)
BODY, MAIN ASSEMBLY, MK 4 MOD 0
IN CONTAINER MK 318 MOD 0

UNIT LOAD DATA

DIMENSIONS	155 1/2 L X 28 ¾ X 31 3/8 H
STACKING HEIGHT	29 3/4
GROSS WT	1245 LBS △
CUBE	75 C CU FT
DOT HAZARD CLASSIFICATION	NONE
CONTAINER DRAWING NO	CD 162244

△ DO NOT USE FOR SHIPPING WEIGHT





NOTE

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL HDBK 30(NAVY)

INDEX

ITEM	PAGE
GENERAL NOTES	2
FTL 40 FT TRAILER	2 & 3
LTL 40 FT TRAILER	4 & 5
DETAILS	6 & 7

DISTRIBUTION STATEMENT A
Approved for public release, distribution is unlimited

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY

SIGNATURE — TECHNICAL DIRECTION AGENT (TDA) — NWHC — DATE 9/12/85

SIGNATURE — SEA SYSCOM BY DIRECTION — DATE 9/12/85

ORIGINATOR — DATE 13 Sept '85

NAVAL WEAPONS HANDLING CENTER
WINSTA EARLE, NEW JERSEY

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

GENERAL NOTES

THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE MINE, UNDERWATER, MK 67 2 (SLMM).

2. THE PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS. THE TRAILER AXLES SHALL BE LOCATED IN THE WESTERN HALF OF THE REAR OF THE TRAILER.

3. MKS ARE SHIPPED TO SHIP CONTAINERS TWO HIGH. THEY SHALL BE STACKED AND STRAPPED TOGETHER AS SPECIFIED IN THE CONTAINER UNITIZING DETAIL ON PAGE 5.

4. CHAIN AND STEEL STRAPS MAY BE USED INTERCHANGEABLY FOR TIEDOWNS ON A 1:1 BASIS. THE STRAP USED FOR TIEDOWNS SHALL BE 2 X .050 AND SHALL BE APPLIED AS SPECIFIED AS SHOWN IN THIS DOCUMENT AND THE GENERAL TRUCKLOADING DOCUMENT MIL STD 1320 (NAVY).

5. CHAINS, FITTINGS AND LOAD BINDERS SHALL MEET THE REQUIREMENTS OF AND BE APPLIED AS SPECIFIED BY THE GENERAL TRUCKLOADING DOCUMENT MIL STD 1320 (NAVY) AND THE REQUIREMENTS OF THIS DOCUMENT.

6. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY 49 CFR 392.7 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOADBINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOADBINDERS HAVE BEEN INSPECTED AND SHALL CONCUR IN ITEM NO. 27 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOADBINDER.

8. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR TRAILER WILL NOT EXCEED THESE LIMITATIONS.

9. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO INSIDE OF TRAILER IN AN ACCESSIBLE AREA AND CLOSE AND SEAL TRAILER DOORS.

10. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM L 751; NAILS, FF N 105, TYPE II, STYLE 10, COMMON BRIGHT; STRAPPING, QQ S 781, TYPE I, HEAVY DUTY, FINISH A, B OR C.

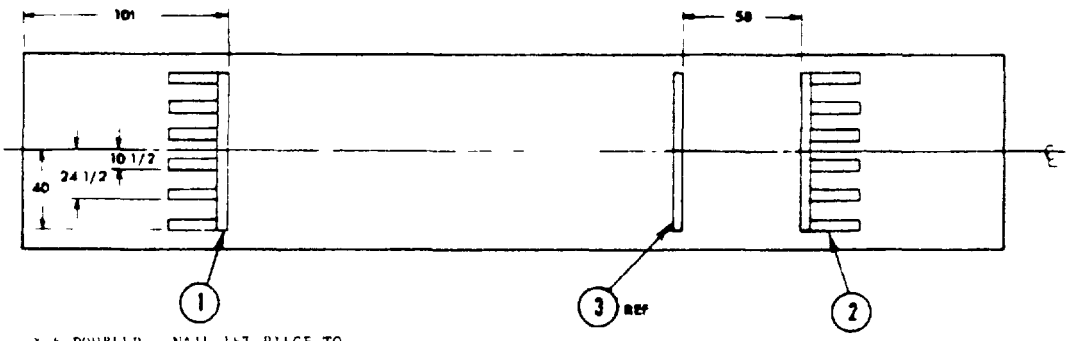
11. FOR GENERAL TRUCKLOADING PROCEDURE REFER TO THE GENERAL TRUCKLOADING DOCUMENT MIL STD 1320 (NAVY).

FTL 40 FT TRAILER (FLATBED)

1. A FULL TRUCKLOAD CONSISTS OF 18 CONTAINERS. SHIPMENT OF CONTAINERS OF MINES IS LIMITED TO TWO HIGH DUE TO THE STACKING CHARACTERISTICS OF THE MK 316 CONTAINER.

2. PREPOSITION BLOCKING AS SHOWN IN DETAIL A. THE SUGGESTED LOADING SEQUENCE IS:

A. INSTALL REAR PREPOSITIONED BLOCKING
B. LOAD REAR STACK OF CONTAINERS
C. LOAD MIDDLE STACK OF CONTAINERS
D. INSTALL FORWARD PREPOSITIONED BLOCKING
E. LOAD FORWARD STACK OF CONTAINERS
F. COMPLETE REMAINING BLOCKING AND TIEDOWN

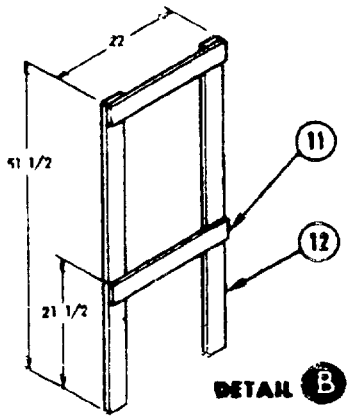


2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR 5/1 16d NAIL EVERY 8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/4 16d NAILS. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

DETAIL A

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

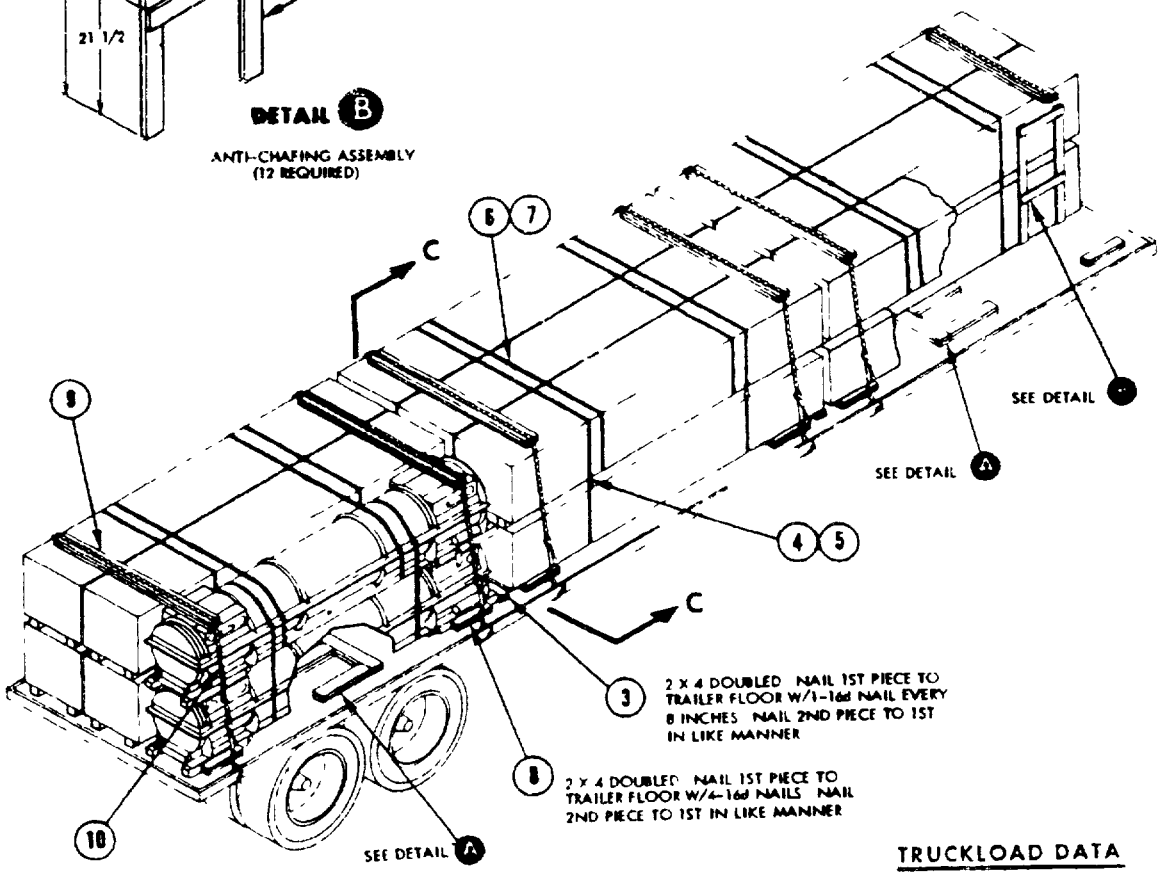


DETAIL B

ANTI-CHAFING ASSEMBLY
(12 REQUIRED)

NOTE

THIS FTL ILLUSTRATES THE USE OF CHAINS AS TIEDOWNS 2 X .050 STEEL STRAPS ARE AN AUTHORIZED ALTERNATE

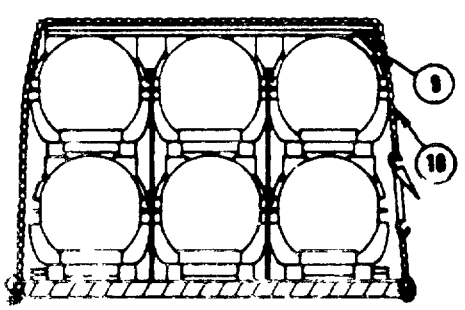


3 — 2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 8 INCHES NAIL 2ND PIECE TO 1ST IN LIKE MANNER

8 — 2 X 4 DOUBLED NAIL 1ST PIECE TO TRAILER FLOOR W/4-16d NAILS NAIL 2ND PIECE TO 1ST IN LIKE MANNER

TRUCKLOAD DATA

NO OF CONTAINERS	18
WEIGHT (APPROX)	22 410 LBS



SECTION C-C

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
12	UPRIGHT	1 X 4 X 51 1/2	24	SEE 11	-	-
11	TIE PIECE	1 X 4 X 22	24	12	3	6d CLINCHED
10	CHAIN & BINDER	5/16 OR 3/8	6	-	-	-
9	PROTECTOR BOARD	2 X 6 X 76	12	DOUBLE LAMINATE	8	10d
8	SLEEPER	2 X 4 X 18	24	SEE FIELD NOTE		
7	SEAL	FOR 1 1/4 STRAP	12			-
6	CROSS STRAP	1 1/4 X .035 X 21 FT	6	-	-	-
5	SEAL	FOR 1 1/4 STRAP	36	-	-	-
4	UNITIZING STRAP	1 1/4 X .035 X 16 FT	18	-		-
3	SEPARATOR	2 X 4 X 80	4	SEE FIELD NOTE		
2	BACKUP CLEAT	2 X 6 X 24	24	SEE DETAIL A		
1	CROSSMEMBER	2 X 6 X 80	4	SEE DETAIL A		

LIST OF MATERIALS & NAILING DATA

FTL 40 FT TRAILER (FLATBED)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

LTL 40 FT TRAILER (FLATBED)

1 WHEN REQUIRED TO SHIP LESS THAN A FULL TRUCKLOAD SELECT THE CORRECT LOAD PATTERN FROM TABLE I POSITION THE CONTAINERS ON THE TRAILER USING THIS PATTERN

2 THE LTL PATTERNS SHOWN FOR 11, 13, 14, 15, 16 AND 17 CONTAINERS REQUIRE PREPOSITIONED BLOCKING AS SHOWN IN DETAIL A, PAGE 2 WHEN CONTAINERS ARE TWO ACROSS MODIFY BLOCKING ACCORDINGLY BLOCK AND TIEDOWN CONTAINERS USING THE PRINCIPLES SHOWN ON PAGE 3 AND LTL NOTES ON PAGE 4

3 WHEN REQUIRED TO SHIP 17 CONTAINERS THE FILLER ASSEMBLY, DETAIL F PAGE 6 SHALL BE USED TO FILL THE VOID IN THE CENTER OF THE MIDDLE STACKS WHERE THE STACK IS ONE CONTAINER HIGH THE ASSEMBLY SHALL BE POSITIONED ON THE ONE HIGH CONTAINER AND STRAPPED TO THE CONTAINER IN SIMILAR MANNER AS SHOWN PER CONTAINER UNITIZING ON PAGE 7

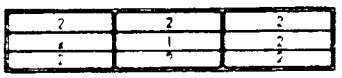
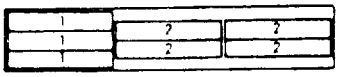
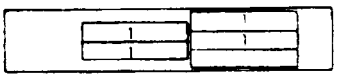
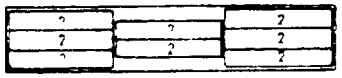
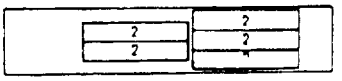
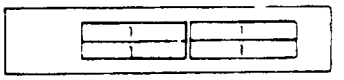
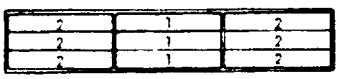
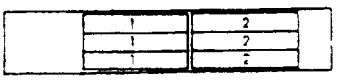
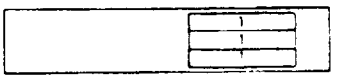
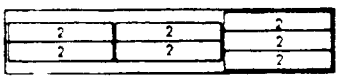
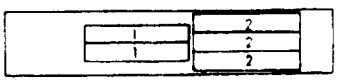
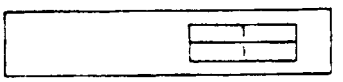
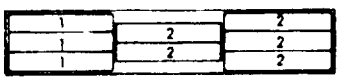
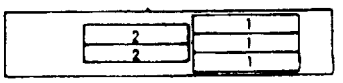
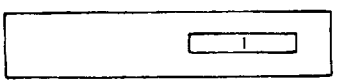
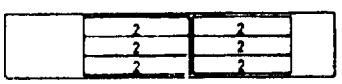
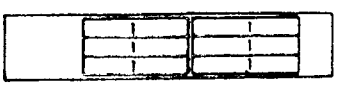
4 THE LTL PATTERN SHOWN FOR 12, 10, 9, 8 & 7 CONTAINERS REQUIRES THE FORWARD STACK TO BE POSITIONED 46 INCHES FROM THE FRONT OF THE TRAILER BLOCK AND TIEDOWN CONTAINERS USING THE PRINCIPLES SHOWN ON PAGE 3 AND 5

5 THE LTL PATTERN SHOWN FOR 6 OR LESS CONTAINERS REQUIRES THE FORWARD STACK TO BE POSITIONED 46 INCHES FROM THE FRONT OF THE TRAILER BLOCK AND TIEDOWN CONTAINERS USING THE PRINCIPLES SHOWN ON PAGE 5

6 THE LTL ON PAGE 5 IS FOR 5 CONTAINERS LTL SHIPMENTS ARE NOT LIMITED TO THIS QUANTITY ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF CONTAINERS TO BE SHIPPED ON A TRAILER AND THEY SHOULD BE BLOCKED AND TIED DOWN IN ACCORDANCE WITH APPROPRIATE PARTS OF THIS STANDARD

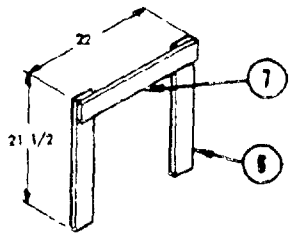
TABLE I

LOAD PATTERN ✱

NO OF CONTAINERS	→ FWD	NO OF CONTAINERS	→ FWD	NO OF CONTAINERS	→ FWD
17		11		5	
16		10		4	
15		9		3	
14		8		2	
13		7		1	
12		6			

✱ LOAD PATTERN IS THE FLOOR PLAN FOR THE LOAD. THE NUMBER IN THE BLOCKS IS THE NUMBER OF CONTAINERS IN THE LAYER OF THE STACK

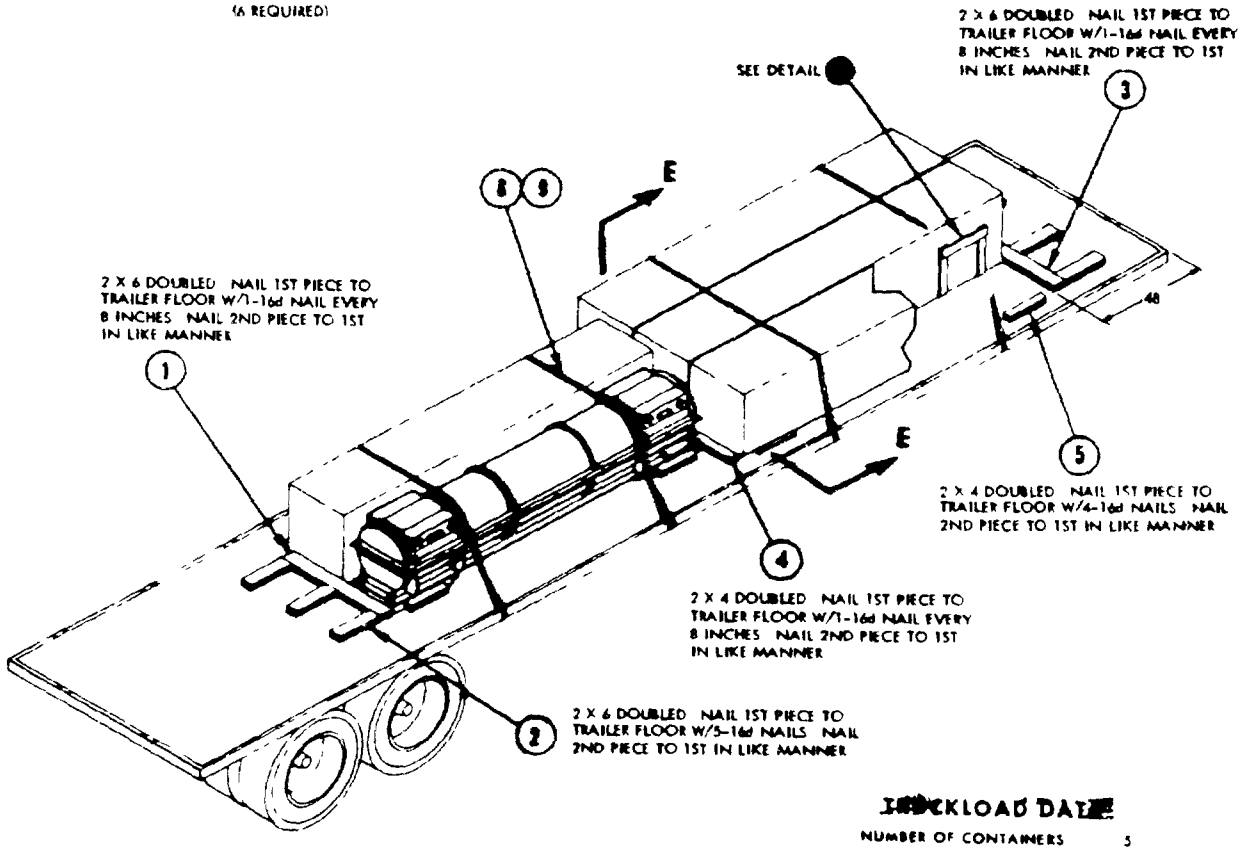
Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

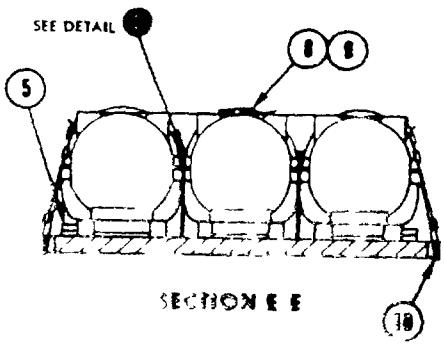


DETAIL D
ANTI CHAFFING ASSEMBLY
(6 REQUIRED)

NOTE

THIS LOAD ILLUSTRATES THE USE OF 2 X .050 STEEL STRAPS AS TIEDOWNS CHAIN AND LOADBINDERS ARE AN AUTHORIZED ALTERNATE





SECTION E E

[illegible]CKLOAD DATA

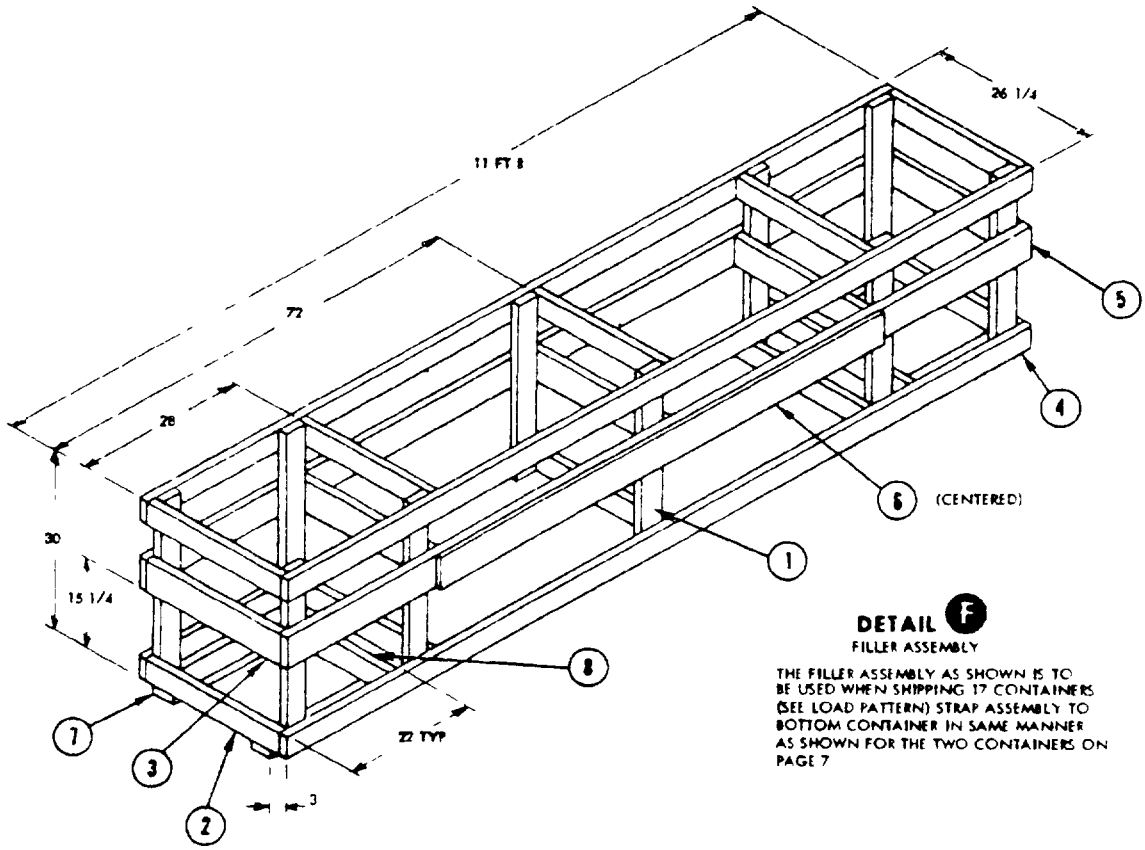
NUMBER OF CONTAINERS 5
WEIGHT (APPROX) 6 225 LBS

PIECE NO	DESCRIPTION	SIZE	NO OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
10	PAD (STRAP)	2 X .050 X 18	8	-	-	
9	SEAL	FOR 2 INCH STRAP	32	-	-	
8	STRAP	2 X .050 X 18 FT 6	4	-	-	-
7	TIE PIECE	1 X 4 X 2'	6	6	7	6d CLINCHED
6	UPRIGHT	1 X 4 X 21 1/2	12	SEE 7	-	
5	SLEEPER	2 X 4 X 18	16	SEE FIELD NOTE		
4	SEPARATOR	2 X 4 X 80	2	SEE FIELD NOTE		
3	CROSSMEMBER	2 X 6 X 80	2	SEE FIELD NOTE		
2	BACKUP CLEAT	2 X 6 X 20	14	SEE FIELD NOTE		
1	CROSSMEMBER	2 X 6 X 50	7	SEE FIELD NOTE		

LIST OF MATERIALS & NAILING DATA

TYPICAL [illegible] 40 FT TRAILER (FLATBED)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.



DETAIL F
FILLER ASSEMBLY

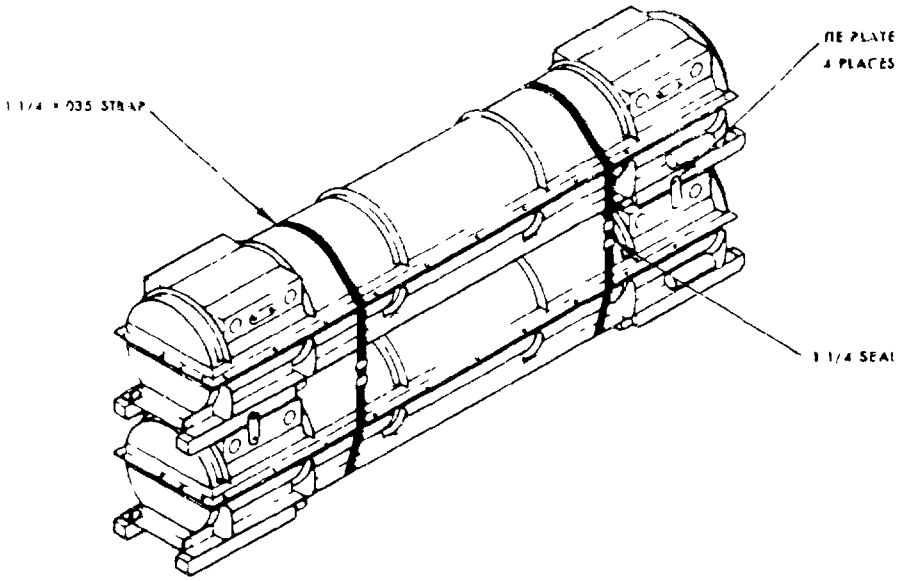
THE FILLER ASSEMBLY AS SHOWN IS TO BE USED WHEN SHIPPING 17 CONTAINERS (SEE LOAD PATTERN) STRAP ASSEMBLY TO BOTTOM CONTAINER IN SAME MANNER AS SHOWN FOR THE TWO CONTAINERS ON PAGE 7

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER NAILS	SIZE
8	SUPPORT BRACE	[illegible] X 4 X 23 1/4	2	SEE 4		
[illegible]	SUPPORT	2 X 4 X 22	4	2, 8	2/JOINT	16d
6	BUFFER PIECE	1 X 6 X 7 FT	2	6	1/FT	10d
5	LONGITUDINAL	[illegible] X 6 X 11 FT 8	[illegible]	[illegible]	1/JOINT	16d
4	LONGITUDINAL	[illegible] X 4 X 11 FT 8	4	8	2/JOINT	16d
				7	1/JOINT	16d
	CROSS BRACE	[illegible] X 6 X 23 1/4	5	1	3/JOINT	16d
2	HORIZONTAL	2 X 4 X 23 1/4	10	1	2/JOINT	16d
1	VERTICAL	2 X 4 X 30	10	4, 5	2/JOINT	10d

LIST OF MATERIALS AND NAILING DATA

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

UNITIZING PROCEDURE



CONTAINER UNITIZING PROCEDURE

1. STACK TWO CONTAINERS USING A FORK LIFT TRUCK
2. BE CERTAIN THAT THE NESTING FEATURES AND BEARING SURFACES ARE PROPERLY POSITIONED TO PREVENT SHIFTING IN LATERAL AND LONGITUDINAL DIRECTIONS
3. FASTEN TIE PLATES IN POSITION ON CONTAINERS
4. SECURE BOTTOM CONTAINER TO TOP CONTAINER WITH TWO 1 1/4 X .035 STEEL STRAPS POSITIONED AS INDICATED ABOVE
5. AFTER TENSIONING STRAPS, SEAL WITH TWO 1 1/4 INCH DOUBLE NOTCHED SEAL

WARNING

WHEN LOADING/UNLOADING TRAILERS WITH UNITIZED CONTAINERS, EXTRA CAUTION SHOULD BE TAKEN TO PREVENT TOPPLING. SPECIAL ATTENTION SHOULD BE GIVEN TO APPROPRIATE BACKUP OF OUTBOARD CONTAINERS AND CONTAINERS IN TOP OUTER ROWS. CONTAINERS SHALL BE DEUNITIZED AFTER UNLOADING THE VEHICLE.

REVIEW ACTIVITY
NAVY OS

PREPARING ACTIVITY
NAVY OS
(PROJECT NO. 8140-N611)

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

U.S. GOVERNMENT PRINTING OFFICE: 1983 - 505-038/21308

INSTRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge (DO NOT STAPLE), and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE. This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Weapons Station Earle (NAVSTA)
ATTN: Code 803 - S. Paquette
Colts Neck, NJ 07722

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1 DOCUMENT NUMBER	2 DOCUMENT TITLE

3a NAME OF SUBMITTING ORGANIZATION	4 TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a NAME OF SUBMITTER *(Last, First, MI)* - Optional	b WORK TELEPHONE NUMBER *(Include Area Code)* - Optional
c MAILING ADDRESS *(Street, City, State, ZIP Code)* - Optional	8 DATE OF SUBMISSION *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:48Z
Check the source to verify that this is the current version before use.